Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Algorithms and Mathematical Concepts for Computing**
Assessment Type	**Examination**
Date	**27th February 2025**
Time	**2.00pm – 5.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

This question relates to the Software Development Process and Algorithm:

a) At which stage of the software development lifecycle is the actual coding and integration of the system components carried out to turn the design into a working application? **1**

Mark Scheme:

Implementation (1 mark)

b) In which stage of the software development lifecycle is the application validated for functionality, usability and security to ensure it meets the defined specifications and user requirements? **1**

Mark Scheme:

Testing (1 mark)

c) Explain the difference between an algorithm and a program. Discuss the purpose of algorithms and programs.
Describe the hierarchy of algorithmic complexity, providing an example for each complexity level. **8**

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Marks

Mark Scheme:

Criteria:

Algorithm vs Program:

Algorithm Definition: An algorithm is a finite set of precise instructions for solving a problem or performing a task. (1 mark)
Program Definition: A program is the implementation of an algorithm in a programming language that can be executed by a computer. (1 mark)
Purpose of Algorithms: The purpose of an algorithm is to provide a systematic, step-by-step procedure to solve a specific problem. (1 mark)
Purpose of Programs: The purpose of a program is to execute an algorithm to automate a task or process, making it operational on a computer system. (1 mark)

Hierarchy of Algorithmic Complexity:

Easy Algorithms: Example: Finding the largest number in a list. These algorithms involve simple operations like comparisons. (1 mark) Any other valid alternative example will be accepted.
Moderately Difficult Algorithms: Example: Sorting a list of numbers, which requires additional steps. (1 mark) Any other valid alternative example will be accepted.
Very Hard Algorithms: Example: Finding the shortest path between distant locations, which involves complex optimisation. (1 mark) Any other valid alternative example will be accepted.
Essentially Impossible Algorithms: Example: Factoring large composite numbers, which is computationally infeasible for classical computers. (1 mark) Any other valid alternative example will be accepted.

Total 10 Marks

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Marks

Question 2

This question relates to numbers and number systems:

a) Which of the following is **NOT** a valid number system? **1**

i) Binary number system
ii) Decimal number system
iii) Fractional number system
iv) Octal number system

Mark Scheme:

iii) Fractional number system (1 mark)

b) Conversion from decimal to _____ involves converting a number from base 10 to base 8. **1**

Mark Scheme:

Octal (1 mark)

c) Describe the concept of least common multiple (LCM) **and** explain its significance in number theory. **2**

Mark Scheme:

- ***The Least Common Multiple (LCM) is the smallest positive integer that is divisible by all the numbers in a given set. (1 mark)***
- ***It is significant in number theory for solving problems related to synchronisation of cycles or events, as well as in operations with fractions. (1 mark)***

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Marks

d) Perform the following conversions **and** show your working.

i) Convert the octal number 54 to binary. Show your working. **2**

Mark Scheme:

- ***5 in octal = 101 in binary***
- ***4 in octal = 100 in binary***
  So, 54 (octal) = 101100 (binary).
  (1 mark for correct answer, 1 mark for working)

ii) Convert the hexadecimal number 1D to decimal. Show your working. **2**

Mark Scheme:

1D = (1 × 16^1) + (13 × 16^0)
= 16 + 13
= 29 (decimal).
(1 mark for correct answer, 1 mark for working)

iii) Convert the decimal number 75 to binary. Show your working. **2**

Mark Scheme:

Convert 75 to binary:

1. ***75÷2=37 div 2 = 37, remainder 1***
2. ***37÷2=18 div 2 = 18, remainder 1***
3. ***18÷2=9 div 2 = 9, remainder 0***
4. ***9÷2=4 div 2 = 4, remainder 1***
5. ***4÷2=2 div 2 = 2, remainder 0***
6. ***2÷2=1 div 2 = 1, remainder 0***
7. ***1÷2=0 div 2 = 0, remainder 1***

Now, write the remainders from bottom to top:

1001011

(1 mark for correct answer, 1 mark for working)

Total 10 Marks

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Marks

Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following operators represents the logical ***AND*** operation? **1**

i) $\wedge$
ii) $\vee$
iii) $\neg$
iv) $\rightarrow$

Mark Scheme:

i) $\wedge$ ***(1 mark)***

b) Complete the truth table for the Boolean expression: **4**

$$(P \vee Q) \wedge \neg (P \wedge Q)$$

P	**Q**	**P ∧ Q**	**P ∨ Q**	**¬(P ∧ Q)**	**(P ∨ Q) ∧ ¬(P ∧ Q)**
T	T				
T	F				
F	T				
F	F				

Mark Scheme:

Accurate completion of the truth table (each correct column = 1 mark)

Answer:

P	**Q**	**P ∧ Q**	**P ∨ Q**	**¬(P ∧ Q)**	**(P ∨ Q) ∧ ¬(P ∧ Q)**
T	T	**T**	**T**	**F**	**F**
T	F	**F**	**T**	**T**	**T**
F	T	**F**	**T**	**T**	**T**
F	F	**F**	**F**	**T**	**F**

Please note that if the result of the third column (P ∨ Q) and/or the fourth column (¬(P ∧ Q)) is incorrect, but the calculation for the fifth column ((P ∨ Q) ∧ ¬(P ∧ Q)) is correct based on the incorrect results of the third and/or fourth column, the mark will still be awarded.

For example, if the first-row result of the third and fourth columns is T and T instead of T and F, and the fifth column result is T based on this, the mark for the fifth column will be awarded if all the row values in the fifth column are correct based on the incorrect or correct results of the third and fourth columns.

c) Explain the concept of logical implication in Boolean algebra **and** provide TWO (2) examples. Include the truth table for logical implication. **5**

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Marks

Mark Scheme:

- ***Concept of logical implication:***
  - ***Logical implication P→Q is true unless P is true and Q is false.***

    1 mark for correctly defining logical implication

  - ***True when P is false or both P and Q are true.***
  - ***False only when P is true and Q is false.***

    1 mark for explaining the truth conditions of logical implication

- ***Truth table for logical implication:***

P	***Q***	***P→Q***
T	***T***	***T***
T	***F***	***F***
F	***T***	***T***
F	***F***	***T***

(1 mark for the correct truth table)

Two examples of logical implication:

- ***Example 1: "If it rains (P), then the ground will be wet (Q)" means that if it is raining (P) and the ground is not wet (Q), the implication is false. Otherwise, it holds true. (1 mark)***
- ***Example 2: "If a student studies (P), then they are more likely to pass the exam (Q)" means that if the student studies (P) and does not pass the exam (Q), the implication is false. Otherwise, it holds true. (1 mark)***

Total 10 Marks

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Marks

Question 4

Answer the following questions related to Predicate Logic, Set Theory and Venn Diagrams:

a) Which of the following symbols represents the intersection operation in set theory? **1**

i) ⊆
ii) ⊂
iii) ∪
iv) ∩

Mark Scheme:

iv) ∩ ***(1 mark)***

b) Explain the concept of set union **and** provide ONE (1) example. **2**

Mark Scheme:

- ***The union of two sets refers to a set containing all elements that belong to either set or both. (1 mark)***
- ***Example:***
  If A={1,2} and B={2,3}, then A ∪B={1,2,3}. (1 mark) Accept valid alternatives.

c) Calculate the complement of set A = {1,3,5} relative to the universal set U = {1,2,3,4,5,6} **and** explain the process. **2**

Mark Scheme:

- ***A' = {2,4,6}.***
  (1 mark for correctly identifying the complement of A)
- ***The complement of a set includes all the elements in the universal set U that are not in A. Therefore, $A' = U - A = \{2,4,6\}$.***

(1 mark for explaining the process)

d) Describe the power set of a set. Calculate the power set of A = {x,y} **and** provide the correct representation. **3**

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Marks

Mark Scheme:

- ***The power set of a set is the set of all subsets, including the empty set and the set itself.***
  (1 mark for explaining the power set)

- ***P(A)={∅,{x},{y},{x,y}}***
  - ***1 mark for correctly listing the subsets: ∅,{x},{y},{x,y}.***
  - ***1 mark for correct formatting and representation of the power set (i.e., using the proper set notation).***

e) Explain the key difference between propositional logic and predicate logic in terms of expressiveness. **2**

Mark Scheme:

- ***Propositional logic deals with simple propositions and logical connectives (1 mark)***
- ***while predicate logic allows representation of relationships and quantification over objects. (1 mark)***

Total 10 Marks

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Marks

Question 5

Answer the following questions related to statistics:

a) Explain the concept of variance. Provide ONE (1) example of a scenario where it is appropriate to use variance. **2**

Mark Scheme:

- ***Variance quantifies how much data points deviate from the mean, representing data spread. Higher variance indicates greater variability, while lower variance suggests data are closer to the mean. (1 mark)***
- ***Example: (1 mark)***
    - ***Appropriate: Analysing financial market volatility.***

b) Write the formula for calculating variance. **3**
Calculate the variance for the following data set: 4, 8, 6, 5, 9. Show your working.

Mark Scheme:

1. ***Formula of Variance: $\Sigma(xi-mean)^2 / n$ (1 mark)***
2. ***Calculate the mean:***
   Mean = (4+8+6+5+9) / 5 = 6.4
   (1 mark)
3. ***Compute variance using $\Sigma(xi-mean)^2 / n$:***
   Variance = $(4-6.4)^2+(8-6.4)^2+(6-6.4)^2+(5-6.4)^2+(9-6.4)^2 / 5 = 3.44$ (1 mark)

c) Describe the concept of kurtosis in statistics **and** explain its importance in analysing data distribution. **2**

Mark Scheme:

- ***Kurtosis measures the "tailedness" of a data distribution, identifying the presence of outliers or the concentration of data around the mean. (1 mark)***
- ***It helps determine whether a dataset has a normal distribution, with applications in financial risk analysis and statistical modelling. (1 mark)***

d) Describe median **and** explain how it is calculated. **3**
Determine the median for the following data set: 12, 7, 15, 10, 5. Show your working.

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Marks

Mark Scheme:

1. ***What is the median: The median is the middle value in a data set arranged in order. For an odd number of values, it's the middle value. For an even number of values, it's the average of the two middle values. (1 mark)***
2. ***How it is calculated: If there is an odd number of data points, the median is the value at the centre. If there is an even number of data points, the median is the average of the two middle values. (1 mark)***
3. ***Median for the following data set: (1 mark)***
    - ***Arrange the data in ascending order:***
      5, 7, 10, 12, 15
    - ***Identify the middle value for an odd number of data points:***
      Median = 10

Total 10 Marks

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Marks

Question 6

Answer the following questions related to Data Structures and Arrays:

a) Which of the following is **NOT** a characteristic of an array? **1**

i) Stores elements of the same data type.
ii) Allows random access to elements.
iii) Can automatically sort its elements.
iv) Elements are stored in contiguous memory locations.

Mark Scheme:

iii) Can automatically sort its elements. (1 mark)

b) Define the data type "queue". Explain TWO (2) operations on a queue and provide ONE (1) example of an application where it would be used. **4**

Mark Scheme:

- ***Definition: A queue is a data structure that follows the First-In-First-Out (FIFO) principle, where the first element added is the first one to be removed. (1 mark)***
- ***Operations: Key operations of a queue include enqueue (adding an element to the back), (1 mark)***
  dequeue (removing the front element) and peek (viewing the front element without removing it). (1 mark)
- ***Application: Queues are used in various applications, such as scheduling processes in operating systems, handling requests in web servers and managing tasks in print spooling. (1 mark: The answer is expected to provide just one application. Any of the given applications or any other valid application will be accepted.)***

c) Write pseudocode to dequeue an element from a queue. Assume the existence of a function called isEmpty(queue) that checks if the queue is empty. **5**

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Marks

Mark Scheme:

Function dequeue(queue):

if not isEmpty(queue): # (1 mark) Using 'not' to check if the queue is not empty

(1 mark) Using the 'isEmpty' function to check if the queue is empty

element = queue[0] # (1 mark) Access the first element in the queue

queue = queue[1:] # (1 mark) Remove the first element by slicing the queue

return element # (1 mark) Return the removed element

End Function

Total 10 Marks

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Marks

Question 7

Answer the following questions related to Data Structures:

a) Which of the following data structures stores elements in nodes where each node contains a reference (or pointer) to the next node in the sequence? **1**

i) Array
ii) Queue
iii) Stack
iv) Linked List

Mark Scheme:

iv) Linked List (1 mark)

b) In a stack, elements are added to the _____ and removed from the _____. **2**

Mark Scheme:

- ***top (1 mark)***
- ***top (1 mark)***

c) Consider a queue initially containing the elements 10, 20, 30, 40. Assume 10 is at the front of the queue. Perform the following operations one after the other: **3**

1. Add the number 50.
2. Dequeue TWO (2) elements.
3. Add the numbers 60 and 70.

Draw out the queue after each operation. Show your working.

Mark Scheme:

Step-by-Step Operations:

1. ***Add 50 → Queue: [10, 20, 30, 40, 50] (1 mark)***
2. ***Dequeue two elements: Remove 10 and 20 → Queue: [30, 40, 50] (1 mark)***
3. ***Add 60 and 70 → Final Queue: [30, 40, 50, 60, 70] (1 mark)***

d) Provide TWO (2) computing-related examples where a stack data structure would be appropriately used. Explain your answer for each example. **4**

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Marks

Mark Scheme:

- ***Example 1: A computing-related example where a stack data structure would be appropriately used is the Undo functionality in a text editor. (1 mark)***
    - ***Each action (e.g., typing, deleting) is pushed onto the stack, and the most recent action can be undone by popping it off the stack. (1 mark)***
- ***Example 2: A computing-related example of a stack is a web browser's navigation history (Back button). (1 mark)***
    - ***Each visited page is pushed onto the stack, and when the user clicks the "Back" button, the most recent page is popped off the stack to navigate to it. (1 mark)***

Note: Accept other valid alternatives, such as recursion or managing function calls in programming.

Total 10 Marks

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Marks

Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following best describes the concept of asymptotic analysis in algorithms? **1**

i) It measures the exact runtime of an algorithm for specific input sizes.
ii) It evaluates the algorithm's performance for very large input sizes.
iii) It calculates the number of operations for the smallest input size.
iv) It determines the memory usage of an algorithm in all cases.

Mark Scheme:

***ii) It evaluates the algorithm's performance for very large input sizes*. (1 mark)**

b) Using the Big-O notation state the worst-case time complexity for a linear search algorithm. **1**

Mark Scheme:

O(n) (1 mark)

c) Write pseudocode for a function that implements an iterative linear search algorithm to find a target value in an array. The function should take the array and target value as inputs and return the index of the target value, or -1 if the target is not found. **6**

Mark Scheme:

```
Function linearSearch(arr, target):
    for i = 0 to length(arr) - 1:  # Loop through each element in the array
        if arr[i] == target:       # Check if the current element equals the target
            return i               # Return the index if the target is found
    return -1                      # Return -1 if the target is not found
End Function
```

Criteria:

- ***1 mark for correct parameters (arr and target).***
- ***1 mark for correct loop setup (looping through each element in the array).***
- ***1 mark for checking if the current element equals the target.***
- ***1 mark for correctly returning the index when the target is found.***
- ***1 mark for correctly returning -1 if the target is not found after the loop.***
- ***1 mark for proper function structure (having the function name, parameters and correct ending).***

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Marks

d) Explain the efficiency of binary search with respect to its time complexity. Provide ONE (1) scenario where binary search is more suitable than other search algorithms. **2**

Mark Scheme:

- ***Efficiency of Binary Search:***
  Binary search is more efficient for large, sorted datasets because it repeatedly divides the search space in half, resulting in a time complexity of O(log n). (1 mark)
- ***Scenario for Binary Search:***
  A scenario where binary search is more suitable is searching for a value in a sorted list or array, such as looking up a word in a dictionary. (1 mark)

Total 10 Marks

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Marks

Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which sorting algorithm repeatedly divides the input array into smaller subarrays, sorts each subarray and then combines them to produce the final sorted array? **1**

i) Bubble Sort
ii) Quick Sort
iii) Merge Sort
iv) Selection Sort

Mark Scheme:

***iii) Merge Sort* (1 mark)**

b) State what is meant by a recursive base case in programming **and** explain its importance in ensuring the termination of a recursive function. **3**

Provide ONE (1) example scenario where the base case is crucial.

Mark Scheme:

- ***A recursive base case is a condition that stops the recursive function from calling itself further (1 mark).***
- ***It provides a simple and direct solution for the smallest subproblem, ensuring the recursion terminates and prevents infinite recursion (1 mark).***
- ***Example scenario: In calculating the factorial of a number, the base case is factorial (1) = 1, which ensures the recursion stops and prevents an infinite loop. (1 mark) Any other valid alternative example will be accepted.***

c) Fill in the blanks in the pseudocode for computing the sum of numbers from 1 to n using recursion: **4**

```
Function sumToN(n):
  if _____:
     return _____
  return _____ + _____ (n - 1)
End function
```


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Marks

Mark Scheme:

```
Function sumToN(n):
  if n == 1:
     return 1
  return n + sumToN(n - 1)
End function
```

- ***1 mark for correctly completing the base case check: if n == 1:***
- ***1 mark for correctly returning the value when the base case is met: return 1***
- ***1 mark for correctly handling the recursive addition: return n +***
- ***1 mark for correctly completing the recursive call to the function: sumToN(n - 1)***

d) Explain how recursion can be used to solve the Tower of Hanoi problem. **2**

Mark Scheme:

- ***Recursion is used to break the Tower of Hanoi problem into smaller subproblems by moving n-1 disks to an auxiliary peg, then moving the nth disk to the target peg (1 mark).***
- ***Each function call manages a smaller subset of the disks until only one disk remains to be moved, at which point the recursion starts unwinding (1 mark).***

Total 10 Marks

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Marks

Question 10

Answer the following questions related to tree data structures:

a) Which of the following is ***TRUE*** about a binary search tree (BST)? **1**

i) The left subtree of a node contains only nodes with values greater than the node.
ii) The right subtree of a node contains only nodes with values smaller than the node.
iii) For each node, its left and right subtrees are binary search trees.
iv) The root of the tree contains the smallest value in the entire tree.

Mark Scheme:

- **iii)** ***For each node, its left and right subtrees are binary search trees. (1 mark)***

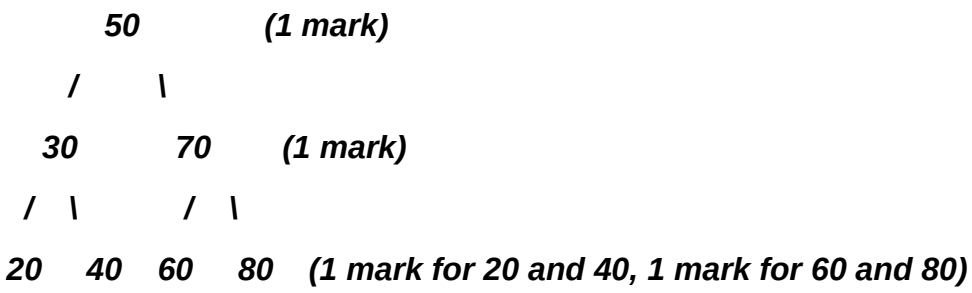
b) In a balanced binary tree, the height difference between the left and right subtrees of any node is at most _____. **1**

Mark Scheme:

1 (1 mark)

c) Construct the binary search tree (BST) for the following sequence of numbers: 50, 30, 20, 70, 40, 80, 60. Ensure that 50 is the root. **4**

Mark Scheme:



d) Using the binary search tree (BST) constructed in Question c), explain the steps to search for the value 25. Describe each comparison and movement until the value is found or determined to be absent. **4**

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Marks

Mark Scheme:

1. ***Start at the root node (50):***
    - ***Compare 25 with 50: 25 < 50, so move to the left subtree (30).***
    - ***(1 mark)***
2. ***Check node (30):***
    - ***Compare 25 with 30: 25 < 30, so move to the left subtree (20).***
    - ***(1 mark)***
3. ***Check node (20):***
    - ***Compare 25 with 20: 25 > 20, so move to the right subtree.***
    - ***(1 mark)***
4. ***Check node (null):***
    - ***The right subtree of node 20 is empty, meaning the value 25 is not found.***
    - ***(1 mark)***

Please note that if the answer for the tree constructed in c) is incorrect, but the explanation for searching 25 is correct relative to the constructed tree in c), the student will still receive full marks.

Total 10 Marks

End of paper

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Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO1	Yes
2	LO5	Yes
3	LO2, LO3	Yes
4	LO4, LO6	Yes
5	LO9	Yes
6	LO7	Yes
7	LO7	Yes
8	LO8	Yes
9	LO8	Yes
10	LO7, LO8	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Explain and use Computational Thinking	Demonstrates no understanding of Computational Thinking and its role.	Provides a basic explanation of Computational Thinking.	Discusses specific examples of Computational Thinking in various contexts.	Applies Computational Thinking to solve complex problems effectively.	Provides critical analysis and innovative approaches to Computational Thinking.
Explain and apply Boolean algebra concepts and notation.	Unable to explain Boolean algebra concepts and notation.	Provides a basic explanation of Boolean algebra concepts and notation.	Draws logic diagrams and writes Boolean expressions accurately.	Applies Boolean algebra concepts proficiently in various computing scenarios.	Offers critical analysis and optimizations of Boolean algebra operations.
Explain and use propositional logic concepts and notation	Demonstrates no understanding of propositional logic concepts and notation.	Provides a basic explanation of propositional logic concepts and notation.	Uses truth tables proficiently to evaluate propositions.	Applies propositional logic effectively in analysing complex statements.	Offers critical evaluation of propositional logic limitations and implications.
Explain and use simple predicate logic statements	Unable to explain simple predicate logic statements.	Provides a basic explanation of simple predicate logic statements.	Writes and interprets predicate logic statements involving quantifiers accurately.	Applies predicate logic statements effectively in real-world scenarios.	Provides innovative applications and thorough analysis of predicate logic statements.
Understand and explain number based system	Unable to understand or explain number-based systems.	Provides a basic explanation of number-based systems.	Performs conversions and arithmetic operations accurately in different number bases.	Applies number-based systems proficiently in solving computational problems.	Offers advanced insights and applications of number-based systems.

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Explain and use set theory and notation	Unable to explain set theory concepts and notation.	Provides a basic explanation of set theory and notation.	Uses set notation proficiently to perform operations and interpret Venn diagrams.	Applies set theory concepts effectively in analysing complex relationships.	Offers advanced interpretations and applications of set theory concepts.
Identify and use common data structures	Demonstrates no understanding of common data structures.	Provides a basic explanation of common data structures.	Applies common data structures proficiently in solving computational problems.	Demonstrates advanced understanding and optimization of data structure usage.	Offers innovative applications and optimizations of common data structures.
Explain and use common algorithms	Unable to explain common algorithms for searching & sorting.	Provides a basic explanation of common algorithms for searching & sorting.	Explains efficiency criteria used to evaluate algorithms effectively.	Applies common algorithms proficiently in solving computational problems.	Offers critical analysis and optimizations of algorithm efficiency.
Understand the fundamental concepts on statistics	Unable to understand or explain key statistical measures and concepts.	Provides a basic explanation of key statistical measures and concepts.	Calculates and uses descriptive statistics effectively.	Applies statistical concepts proficiently in analysing real-world data.	Offers advanced insights and applications of statistical concepts

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